

Aaroh Bhardwaj

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Summary

Backend-focused engineer building reliable, low-latency systems and scalable APIs. Experience with event-driven architectures, data pipelines, and production RAG systems with strong emphasis on correctness, observability, and performance.

Skills

- **Languages:** JavaScript, TypeScript, Python, SQL, C++
- **Backend:** Node.js, Fastify, NestJS, Express, REST APIs, JWT
- **Databases:** PostgreSQL, MongoDB, Redis (Streams, Pub/Sub)
- **Messaging and Realtime:** RabbitMQ, SSE

Work Experience

Full Stack Developer Intern

Sep 2025 – Jan 2026

Phyniks Dreamlabs LLP, Bangalore

- Built and optimized NestJS backends with validation layers, guards, caching, and efficient query patterns, improving API latency and stability.
- Developed SSR Next.js admin dashboards with fast initial load times and consistent server-side data fetching.
- Designed PostgreSQL and MongoDB schemas with indexing (40% faster queries) and access controls, enabling reliable multi-admin workflows.
- Introduced Redis caching, background jobs, and audit logging, reducing database load and improving system resilience.

Projects

Invar — Distributed Metrics Ingestion Platform — [Live] — [Repo]

Dec 2025

- Built a high-throughput ingestion pipeline (1K+ events/min) using Redis Streams with consumer groups and horizontal scaling.
- Optimized cold-path writes to PostgreSQL (700–800 rows/min), achieving **2.5× faster** DB throughput via concurrent batch processing.
- Implemented PEL-based retries, DLQs, batching, and idempotent processing, maintaining near-zero worker backlog under sustained load.
- Delivered real-time dashboards via SSE with near real-time event propagation.

SentinelRAG — Retrieval-Augmented Backend System — [Live] — [Repo]

Jan 2026

- Built a production RAG pipeline with hybrid retrieval (BM25 + embeddings) and LLM reranking.
- Achieved sub-1s retrieval/rerank and low single-digit seconds end-to-end latency.
- Designed ingestion pipeline (parse → chunk → embed → index) for 50–100 page PDFs with stage-wise observability.
- Implemented caching, async workers, and structured logging for predictable performance.
- Structured the system as decoupled services (API, retrieval, worker) to isolate compute-heavy tasks.
- Deployed containerized services on GCP Cloud Run with CI/CD via Cloud Build for repeatable releases.

NexusFeed — Event-Driven Microservices Platform — [Live] — [Repo]

Nov 2025

- Architected a 6-service system with API Gateway, Identity, Posts, Media, and Search services.
- Implemented JWT auth with refresh rotation, rate limiting, and centralized request validation at the gateway.
- Designed RabbitMQ pipelines for async processing and eventual consistency across services.
- Built Docker-based deployment with service isolation, health checks, and internal communication.
- Optimized queries and caching layers to improve feed load performance and responsiveness.

Education

B.E. Computer Science Engineering

2022 – 2026 (Expected)

Acharya Institute of Technology, Bengaluru